

Genetic report for longevity

Nails, Wound Healing and Skin Inflammation

Psoriasis, eczema, rosacea, keloids and healing, hidradenitis, nails and vitiligo

Sérgio F · Report date: 02-10-2026

The inflammatory diseases of the skin are visible inflammation: the same immunity that ages in the rest of the body. This report reads the genes of psoriasis (HLA-C*06:02, IL23R, IL12B, TYK2), atopic eczema, rosacea, healing and keloids, suppurative hidradenitis, nails (fungal infections, nail-wound) and vitiligo and other autoimmune dermatoses.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

Contents

• Summary	3
• Psoriasis	4
• Eczema and atopic dermatitis	6
• Rosacea	8
• Wound healing, keloids and hidradenitis	9
• Nails	11
• Vitiligo and other autoimmune dermatoses	12
• Mechanisms of ageing	13
• Practical part	14
• Limitations	16
• Methods and sources	17

Summary

31

variants

21

in the chip

6

imputed
(DR2 $\geq$ 0.8)

4

not found

6

Scores with
percentile

Overall reading. Skin inflammation has favourable genetics in the major themes: psoriasis at the 6 and 18 percentiles (two PGS), without markers of HLA-C*06:02 and with ERAP1 in favor (CARD14 AA and IFIH1 GG against, small effects); psoriatic arthritis at the 40th percentile; atopic eczema at the 10th percentile and allergic disease in the 22; urticaria at the 3rd percentile. The rosacea (the 31st percentile) and the keloids (the 26th percentile; low transferability loci) are typical, and Dupuytren's disease also (the 38th percentile). The signs against are three: vitiligo at the 90th percentile (48 variants), consistent with the TYR 402R/R, which protects from melanoma but increases vitiligo; suppurative hidradenitis at the 85th percentile (10 loci, weak evidence); and the score of nail diseases at percentile 99.7, but with only 11 loci of a single study of clinical phenotypes (entangled nail, onychomycosis), so it should be read as a weak signal. The score of wounds that do not heal could not be calculated. In practice, healing depends more on rapamycin and isotretinoin than on genes.

Favourable

- **Psoriasis (264 variants):** the 18th percentile
- **Urticaria:** the 3rd percentile
- **Psoriasis (PGS002293, 62 variants):** the 6th percentile (profile §18/§20, imputed genome)
- **Atopic eczema (PGS003486):** the 10th percentile (profile §18/§20)
- **Allergic disease: asthma, rhinitis or eczema (PGS000810):** the 22nd percentile (profile §18/§20)
- **ERAP1 rs27524 GG:** ERAP1: cutting peptides for class I HLA; the effect on psoriasis depends on HLA-C*06:02.
- **EMSY/LRRC32 rs7927894 CC:** Locus 11q13.5: T allele is the most replicated eczema risk outside FLG.
- **CLEC7A rs16910526 AA:** Y238X (Dectin-1): the rare allele truncates the β -glucans receptor; it is associated with onychomycosis and mucocutaneous candidiasis.
- **KIF3A rs6879672 GG:** KIF3A/IL13 (5q31): region of cytokines type 2.

Unfavourable

- **Suppurative hidradenitis:** the 85th percentile
- **Diseases of the nails (entangled nail, onychomycosis, others):** the 100th percentile
- **Vitiligo (PGS000738):** the 90th percentile (profile §18/§20, imputed genome)
- **TYR rs1126809 GG:** R402Q: here the direction is inverse to melanoma: G allele (402R) increases the risk of vitiligo and A (402Q) protects.
- **CARD14 rs11653893 AA:** CARD14: NF- κ B signal in keratinocytes; rare mutations cause pustular psoriasis.
- **IFIH1 rs2111485 GG:** IFIH1 (MDA5): viral RNA sensor.
- **OVOL1 rs10791824 GG:** OVOL1: epidermis differentiation.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 21 chip genotypes, 0 needed complement, 1 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Psoriasis

IL-23/IL-17 axis (IL23R, IL12B, TYK2, TRAF3IP2), antigen presentation (HLA-C*06:02, ERAP1), barrier (LCE3B/C) and NF-κB signal (TNFAIP3). Psoriasis also affects the nails (peaked, onicolysis) in half of the patients.

Ageing mechanisms: Chronic Inflammation; Inflammaging; Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Psoriasis (264 variants)	Favourable	the 18th percentile · z=-0.82	Higher score : higher risk. PGS001871 (PGS Catalog; Penalized regression; Psoriasis; PGP000263, PMID 34995502). Variants with genotype: 261 of 264 (covering of 99.3% coefficients).
Psoriasis (PGS002293, 62 variants)	Favourable	the 6th percentile	Significant genomic loci score. Source: \$18/\$20 profile, imputed genome.
Psoriatic arthritis (PGS000198)	Typical / intermediate	the 40th percentile	Source: \$18/\$20 profile.

Favourable

- Psoriasis (264 variants) (the 18th percentile)
- Psoriasis (PGS002293, 62 variants) (the 6th percentile)
- ERAP1 rs27524 GG

Unfavourable

- CARD14 rs11653893 AA
- IFIH1 rs2111485 GG

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ERAP1	rs27524 5:96766240 · A/C/G/T	GG	chip	A (0; 37%)	moderate	ERAP1: cutting peptides for class I HLA; the effect on psoriasis depends on HLA-C*06:02. Meaning: GWAS Catalog: Psoriasis (Strange A 2010; p=3e-11); Allele A ↑

Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CARD14	rs11653893 17:80205031 · A/G	AA	imputation DR2=1.00	A (2; 49%)	moderate	CARD14: NF-κB signal in keratinocytes; rare mutations cause pustular psoriasis. Meaning: GWAS Catalog: Psoriasis vulgaris (PheCode 696.41) (Verma A 2024; p=3e-14); allele A ↑
IFIH1	rs2111485 2:162254026 · A/G	GG	chip	G (2; 60%)	moderate	IFIH1 (MDA5): viral RNA sensor. Meaning: GWAS Catalog: Psoriatic arthritis (Soomro M 2022; p=1e-20); G ↑ allele

Typical / intermediate variants (7)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HLA-C	rs10484554 6:31306778 · C/T	CC	chip	T (0; 14%)	strong	HLA-C*06:02 tag (the largest genetic factor of psoriasis, OR ~3-4 per copy): the T allele marks the C*06:02.
HLA-C	rs12191877 6:31285148 · C/T	CC	chip	T (0; 14%)	strong	According to the HLA-C*06:02 tag: the T allele marks the risk allele.

IL23R	rs11209026 1:67240275 · G/A	GG	chip	G (2; 94%)	strong	R381Q: A allele (381Q) reduces the IL-23 signal and protects from psoriasis, Crohn's disease and spondylitis; G allele is common (without protection).
TYK2	rs34536443 19:10352442 · G/C	GG	chip	G (2; 97%)	strong	P1104A: the C allele (1104A) greatly reduces the activity of TYK2 (target of deucravacitinib) and protects from psoriasis and other autoimmune diseases; the G allele is the common.
TNFAIP3	rs610604 6:137878280 · G/A/C/T	TG	chip	G (1; 34%)	moderate	TNFAIP3 (A20): NF-κB brake. Meaning: GWAS Catalog: Psoriasis vulgaris (Stuart PE 2015; p=5e-17); G ↑ allele
TRAF3IP2	rs33980500 6:111592059 · C/A/T	CC	chip	T (0; 8%)	moderate	D10N (ACT1): T allele alters the IL-17 signal and increases the risk of psoriasis and psoriatic arthritis.
TNIP1	rs2233278 5:151087628 · G/C/T	GG	chip	C (0; 7%)	moderate	TNIP1: NF-κB brake. Meaning: GWAS Catalog: Psoriasis (Tsoi LC 2012; p=2e-42); allele C ↑

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IL12B	rs3212227 5:159315942 · T/A/G	GT	chip	—	moderate	3'UTR of IL12B (p40, common to IL-12 and IL-23; target of ustekinumab).
LCE3B/ LCE3C	rs4112788 1:152578800 · A/G/T	AA	chip	—	moderate	Tag of LCE3B/LCE3C deletion (epidermal barrier proteins), risk factor of psoriasis.

Eczema and atopic dermatitis

Barriers (FLG, see report Skin and Skin Care) and type 2 immunity (IL13, IL4R, TSLP, EMSY/LRRC32, IL18R1, OVOL1). Scores of eczema and allergic disease calculated before.

Ageing mechanisms: Chronic Inflammation; Intestinal Barrier dysfunction; Immunosenescence

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Atopic eczema (PGS003486)	Favourable	the 10th percentile	Source: \$18/\$20 profile.
Allergic disease: asthma, rhinitis or eczema (PGS000810)	Favourable	the 22nd percentile	Source: \$18/\$20 profile.

Favourable

- Atopic eczema (PGS003486) (the 10th percentile)
- Allergic disease: asthma, rhinitis or eczema (PGS000810) (the 22nd percentile)
- EMSY/LRRC32 rs7927894 CC
- KIF3A rs6879672 GG

Unfavourable

- OVOL1 rs10791824 GG

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
EMSY/LRRC32	rs7927894 11:76590272 · C/T	CC	chip	T (0; 37%)	moderate	Locus 11q13.5: T allele is the most replicated eczema risk outside FLG.
KIF3A	rs6879672 5:132690255 · A/C/G/T	GG	imputation DR2=0.99	A (0; 26%)	moderate	KIF3A/IL13 (5q31): region of cytokines type 2. Meaning: GWAS Catalog: Atopic dermatitis (Chen Y 2022; p=6e-23); Allele A ↑

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
OVOL1	rs10791824 11:65791795 · A/C/G/T	GG	chip	G (2; 57%)	moderate	OVOL1: epidermis differentiation. Meaning: GWAS Catalog: Atopic dermatitis (Budunelis A 2023; p=1e-105); G ↑ allele

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IL13	rs20541 5:132660272 · A/C/G/T	GG	chip	A (0; 21%)	moderate	R130Q: Allele A (130Q) increases the activity of IL-13 (2-type immunity) and the risk of eczema and asthma.
IL18R1	rs72823641 2:102319699 · T/A/C	TT	imputation DR2=1.00	T (2; 86%)	moderate	IL18R1/IL1RL1: 2 type immunity and alarmins. Meaning: GWAS Catalog: Allergic disease (asthma, hay fever or eczema) (Johansson A 2019; p=1e-78); T ↑ allele
RORA	rs10519068 15:60776505 · G/A	GG	chip	G (2; 86%)	moderate	RORA: 2 type innate lymphoid cells. Meaning: GWAS Catalog: Allergic disease (asthma, hay fever or eczema) (Johansson A 2019; p=3e-24); G ↑ allele

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IL4R	rs1805010 16:27344882 · A/C/G/T	AA	chip	—	weak	I50V (IL-4 receiver, dupilumab target): weak associations with atopy. Information.

Rosacea

The rosacea joins innate inflammation, reactive facial vessels and Demodex. GWAS loci: HLA-DRA and BTNL2 (class II of HLA).

Ageing mechanisms: Chronic Inflammation; Vascular Aging

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Rosacea	Typical / intermediate	the 30th percentile · z=-0.54	Higher score : higher risk. Few loci. Loci score of GWAS Catalog ($p\leq 510-8$; 1 variant by 500 kb window; ln(OR) or beta coefficients): 21 of 24 loci with genotype. Studies: Point JL (Rosacea symptom Severity); Verma A (Rosacea (PheCode 695.3)). Variants with genotype: 21 of 24 (covering of 63.7% coefficients).

Favourable

None.

Unfavourable

None.

SNP not found in the data (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HLA-DRA	— — · —	—	SNP not found in the data	—	moderate	HLA-DRA: stronger locus of rosacea. No major variant of GWAS Catalog for this gene with usable genotype and sense.
BTNL2	— — · —	—	SNP not found in the data	—	moderate	BTNL2 (class II of HLA). No major variant of GWAS Catalog for this gene with usable genotype and sense.

Wound healing, keloids and hidradenitis

Keloids are scars that grow beyond the wound (excess of TGF- β and collagen). Suppurative hidradenitis is an inflammatory disease of the follicle in friction zones. Loci: NEDD4 and others (keloids), SOX9 and KLF5 (hydradenite).

Ageing mechanisms: Remodelling of the extracellular matrix; Chronic Inflammation; Cell Senescence; Exhaustion of Stem Cells

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Keloids	Typical / intermediate	the 26th percentile · z=-0.51	Higher score : higher risk. There are few studies; low transferability. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 21 of 26 loci with genotype. Studies: Greene CA (Keloid); Nakashima M (Keloid); Verma A (Keloid scar (PheCode 701.4)). Variants with genotype: 21 of 26 (covering of 80.8% coefficients).
Suppurative hidradenitis	Unfavourable	the 85th percentile · z=1.24	Higher score : higher risk. Loci score of GWAS Catalog ($p \leq 510-8$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 10 of 17 loci with genotype. Studies: Broadway KA (Hidradenitis suppurative); Peaker BL (Hidradenitis suppurative); Verma A (Hidradenitis (PheCode 705.3)). Variants with genotype: 10 of 17 (covering of 58.8% coefficients).
Wounds that do not heal or infect	SNP not found in the data	—	Higher score : higher risk. Very few loci; very low reliability.
Dupuytren's disease (PGS001880)	Typical / intermediate	the 38th percentile	Fibromatosis: example of excessive healing of the fascia. Source: Extracellular Matrix and Glycation report.

Favourable

None.

Unfavourable

- Suppurative hidradenitis (the 85th percentile)

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TNF	rs1800629 6:31575254 · G/A	GG	chip	A (0; 13%)	weak	-308G>A: allele A was associated with more TNF- α and more inflammatory healing in small studies.
NEDD4	rs11632096 15:55918301 · A/G	AG	imputation DR2=0.99	G (1; 28%)	weak	NEDD4: ubiquitin-ligase of the TGF- β pathway; locus of keloids. Meaning: GWAS Catalog: Keloid (Green CA 2025; $p=1e-27$); Allele A ↓

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TGFB1	rs1800470 19:41353016 · G/A/C	AG	chip	—	weak	L10P: the Pro allele (G in strand +) was associated with more TGF- β 1; sense of effect on scars and inconsistent fibrosis. Information.

SNP not found in the data (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SOX9	—	—	—	—	moderate	

	— — · —		SNP not found in the data			SOX9: follicle stem cells; hidradenitis locus. No major variant of GWAS Catalog for this gene with usable genotype and sense.
KLF5	— — · —	—	SNP not found in the data	—	moderate	KLF5: locus of hidradenitis. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Nails

Fungal infections of nails (Dectin-1/CLEC7A, CARD9), nail ingrown and nail diseases in general. The genetics of the nails is poorly studied; the scores have few loci.

Ageing mechanisms: Immunosenescence; Dysbiosis; Remodelling of the extracellular matrix

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Diseases of the nails (entangled nail, onychomycosis, others)	Unfavourable	the 100th percentile · z=3.33	Higher score : higher risk. Few loci. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; $\ln(\text{OR})$ or beta coefficients): 11 of 12 loci with genotype. Studies: Verma A (Dermatophytosis of nail (PheCode 110.11)); Verma A (Diseases of nail, NOS (PheCode 703)); Verma A (Ingrowing nail (PheCode 703.1)). Variants with genotype: 11 of 12 (covering of 93.4% coefficients).

Favourable

- CLEC7A rs16910526 AA

Unfavourable

- Nail diseases (entangled nail, onychomycosis, others) (the 100th percentile)

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CLEC7A	rs16910526 · 12:10118488 · A/C/G	AA	chip	C (0; 7%)	moderate	Y238X (Dectin-1): the rare allele truncates the β -glucans receptor; it is associated with onychomycosis and mucocutaneous candidiasis.

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CARD9	rs4077515 · 9:136372044 · C/A/T	TT	chip	—	weak	S12N: T allele is associated with the strongest CARD9 signal; CARD9 deficiency (rare) causes severe fungal infections. Information.

Vitiligo and other autoimmune dermatoses

In vitiligo, T lymphocytes destroy melanocytes; many loci are the same as pigmentation but with an inverse direction (TYR 402R protects from melanoma but increases vitiligo). Also lichen planus and urticaria.

Ageing mechanisms: Chronic Inflammation; Immunosenescence; Oxidative stress

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Liquen plan	SNP not found in the data	—	Higher score = more risk.
Urticaria	Favourable	the 3rd percentile · z=-1.91	Higher score : higher risk. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 19 of 23 loci with genotype. Studies: Chang D (Chronic spontaneous urticaria); Cornejo-Garcia JA (Acute urticaria and angioedema (non-steroidal anti-inflammatory drug-induced); Kristjansson RP (ICD10 L50: Urticaria); McSweeney SM (Urticaria); Nogawa S (COVID-19 BNT162b1 vaccine 2nd dose adverse event) Sakaue S (Urticaria); Verma A (Urticaria (PheCode 947)); Zhang L (Chronic spontaneous urticaria). Variants with genotype: 19 of 23 (covering of 82.6% coefficients).
Vitiligo (PGS000738)	Unfavourable	the 90th percentile	48 variants. Source: \$18/\$20 profile, imputed genome.

Favourable

- Urticaria (the 3rd percentile)

Unfavourable

- Vitiligo (PGS000738) (the 90th percentile)
- TYR rs1126809 GG

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TYR	rs1126809 · 11:89284793 · G/A	GG	imputation · DR2=1.00	G (2; 75%)	moderate	R402Q: here the direction is inverse to melanoma: G allele (402R) increases the risk of vitiligo and A (402Q) protects.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PTPN22	rs2476601 · 1:113834946 · A/G/T	GG	chip	A (0; 9%)	strong	R620W: Allele A (620W) increases the risk of several autoimmune diseases (vitiligo, diabetes type 1, rheumatoid arthritis, thyroiditis).

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NLRP1	rs2670660 · 17:5615686 · A/C/G	AA	imputation · DR2=0.99	—	moderate	NLRP1: inflammasome; vitiligo locus and associated autoimmune diseases.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Chronic Inflammation	CARD14 AA; IFIH1 GG; OVOL1 GG; Suppurative hidradenitis (the 85th percentile); Vitiligo (PGS000738) (the 90th percentile); TYR GG	Psoriasis (264 variants) (the 18th percentile); Psoriasis (PGS002293, 62 variants) (the 6th percentile); ERAP1 GG; Atopic eczema (PGS003486) (the 10th percentile); Allergic disease: asthma, rhinitis or eczema (PGS000810) (the 22nd percentile); EMSY/LRRC32 CC; KIF3A GG; Urticaria (the 3rd percentile)	Psoriatic arthritis (PGS000198) (the 40th percentile); HLA-C CC; IL23R GG; TYK2 GG; TNFAIP3 TG; TRAF3IP2 CC; TNIP1 GG; IL13 GG
Immunosenescence	OVOL1 GG; Nail diseases (entangled nail, onychomycosis, others) (the 100th percentile); Vitiligo (PGS000738) (the 90th percentile); TYR GG	Atopic eczema (PGS003486) (the 10th percentile); Allergic disease: asthma, rhinitis or eczema (PGS000810) (the 22nd percentile); EMSY/LRRC32 CC; KIF3A GG; CLEC7A AA; Urticaria (the 3rd percentile)	IL13 GG; IL18R1 TT; RORA GG; PTPN22 GG
Intercellular Communication	CARD14 AA; IFIH1 GG	Psoriasis (264 variants) (the 18th percentile); Psoriasis (PGS002293, 62 variants) (the 6th percentile); ERAP1 GG	Psoriatic arthritis (PGS000198) (the 40th percentile); HLA-C CC; IL23R GG; TYK2 GG; TNFAIP3 TG; TRAF3IP2 CC; TNIP1 GG
Inflamming	CARD14 AA; IFIH1 GG	Psoriasis (264 variants) (the 18th percentile); Psoriasis (PGS002293, 62 variants) (the 6th percentile); ERAP1 GG	Psoriatic arthritis (PGS000198) (the 40th percentile); HLA-C CC; IL23R GG; TYK2 GG; TNFAIP3 TG; TRAF3IP2 CC; TNIP1 GG
Remodelling of the extracellular matrix	Suppurative hidradenitis (the 85th percentile); Nail diseases (entangled nail, onychomycosis, others) (the 100th percentile)	CLEC7A AA	Keloids (the 26th percentile); Dupuytren's disease (PGS001880) (the 38th percentile)
Oxidative stress	Vitiligo (PGS000738) (the 90th percentile); TYR GG	Urticaria (the 3rd percentile)	PTPN22 GG
Dysbiosis	Nail diseases (entangled nail, onychomycosis, others) (the 100th percentile)	CLEC7A AA	—
Intestinal Barrier dysfunction	OVOL1 GG	Atopic eczema (PGS003486) (the 10th percentile); Allergic disease: asthma, rhinitis or eczema (PGS000810) (the 22nd percentile); EMSY/LRRC32 CC; KIF3A GG	IL13 GG; IL18R1 TT; RORA GG
Exhaustion of Stem Cells	Suppurative hidradenitis (the 85th percentile)	—	Keloids (the 26th percentile); Dupuytren's disease (PGS001880) (the 38th percentile)
Cell Senescence	Suppurative hidradenitis (the 85th percentile)	—	Keloids (the 26th percentile); Dupuytren's disease (PGS001880) (the 38th percentile)
Vascular Aging	—	—	Rosacea (the 30th percentile)

Practical part

Ideas for the barrier, skin inflammation, scars and nails, crossed with the genetic plan v2 and current medication. The current stack remains as it is; everything is suggestion (not applied). Two drugs of the plan move directly with this theme: rapamycin delays healing (dose-dependent effect, greater on days after taking) and isotretinoin makes the skin thinner and drier.

1. Surgery and procedures far from rapamycin

Genetics: Wound healing (NEDD4, TGFB1; mTOR mechanism)

Mark elective surgeries, dental extractions and dermatological procedures in the middle of the 6 cycle weeks ($\geq 2-3$ weeks after taking) and only resume rapamycin with the closed wound (in general 1-2 weeks later). Always tell the surgeon.

V2 genetic plan and medication: Coherent with the plan v2 (pushed rapamycin). • suggestion (not applied)

2. Procedures and isotretinoin

Genetics: Barriers; healing

Surface procedures (non-ablative laser, superficial peelings) seem safe, but ablative laser, dermoabrasion and elective esthetic surgery are left to stop, unless the dermatologist decides. The isotretinoin also makes the skin of the nails and lips more fragile (paronychia, granulomas).

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

3. Scars: silicone gel and sun protection

Genetics: Keroid score (low transferability); Typical dupuytren

Gel or silicone plates 12-24 h/day during 2-3 months in any new scar; massage from the 3 weeks; FPS 50 in the scar during 12 months (avoids hyperpigmentation). If the keloid score is high or there is a personal history: care with piercings and tattoos.

V2 genetic plan and medication: New; low risk. • suggestion (not applied)

4. Emollients and barrier

Genetics: Eczema p10 and allergic disease p22 (favourable); OVOL1 GG

Hydrating with 2x/day ceramides (more important by isotretinoin than by genes); warm and short showers; soap without fragrance.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

5. Nails: protection, collagen and topical antifungals

Genetics: Score of nail diseases; CLEC7A without Y238X; dapagliflozin (fungal infections)

Gloves for water and detergents; hydrate the cuticle and nail. The collagen 20 g, already in the stack, improved brittle nails in a small trial. If onychomycosis arises: topics first (ciclopirox 8% in varnish or efinazol). Oral terbinafine only with normal ALT; oral itraconazole and fluconazole interact with rapamycin (CYP3A4 inhibitors), so they are outside the 2 weeks after taking. Dapagliflozin increases genital fungal infections: dry well the folds.

V2 genetic plan and medication: Coherent with the v2 plan (oral fluconazole only far from rapamycin). • suggestion (not applied)

6. Vitiligo: monitoring depigmentation

Genetics: Vitiligo p90; TYR GG (without the protective allele 402Q)

In the monthly self-examination of the skin, also look for well-defined white spots (hands, face, friction zones). If they arise: early dermatologist; there is effective topical treatment (JAK inhibitors such as ruxolitinib in cream, tacrolimus) and phototherapy. The daily sunscreen avoids contrast and burns in the spots.

V2 genetic plan and medication: New; no cost. • suggestion (not applied)

7. Psoriasis: favourable genetics; lithium to be monitored

Genetics: Psoriasis p6 and p19; CARD14 AA and IFIH1 GG against; ERAP1 in favor

Lithium is a classic trigger of psoriasis in psychiatric doses; with 5 mg the risk is low, but new desquamative plaques (cotoveles, knees, scalp) justify seeing the dermatologist. Omega-3 and telmisartan, already taken, do not worsen.

V2 genetic plan and medication: Coherent with the plan. • suggestion (not applied)

8. Rosacea and redness

Genetics: Rosacea score (small loci)

If persistent redness or papules appears on the face: azelaic acid 15% or ivermectin 1% in cream; avoid triggering (heat, sun, very hot drinks, spice). Tadalafil and telmisartan can cause facial flushing: it is vasodilation, not rosacea.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

9. LDN and skin inflammation

Genetics: — (immunomodulatory mechanism)

LDN 4.5 mg, already taken, has reports and series of benefit cases in lichen planar pilar, Hairey-Hailey and pruritus; weak evidence, but it is a possible positive secondary effect.

V2 genetic plan and medication: The same as the plan. • suggestion (not applied)

Avoiding or with caution

- Elective surgery or extractions on the days following rapamycin. suggestion (not applied)
- Ablative laser, dermoabrasion and elective esthetic surgery during isotretinoin. suggestion (not applied)
- Oral antifungals (itraconazole, fluconazole) with rapamycin; oral terbinafine with high ALT. suggestion (not applied)
- Piercings and tattoos in risk zones appear a tendency for keloids. suggestion (not applied)
- Topical corticosteroids potent for long periods on the face. suggestion (not applied)

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. HLA is read by tag SNPs, not by typing.
- The keloid scores come from few studies, with low transferability; the rosacea, nails, wounds, lichen and urticaria have few loci. Low reliability.
- The psoriasis, eczema and vitiligo scores explain a greater part of the risk, but they are still not diagnosed; the triggers (infections, stress, medications, friction) weigh a lot.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local statistical imputation (with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001871 (PGS Catalog; Penalized regression; Psoriasis; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Nair RP et al. Sequence and haplotype analysis supports HLA-C as the psoriasis susceptibility 1 gene. *Am J Hum Genet* 2006; Tsoi LC et al. Large scale meta-analysis traits genetic architecture for common psoriasis associated variants. *Nat Common* 2017.
- Duerr RH et al. The genome-wide association study identifies IL23R as an inflammatory antibody disease gene. *Science* 2006; Cargill M et al. A large-scale genetic association study checks IL12B and leads to the identification of IL23R as psoriasis-risk genes. *Am J Hum Genet* 2007.
- Dendrou CA et al. Resolving TYK2 locus genotype-to-phenotype differences in autoimmunity. *Sci Transl Med* 2016.
- Paternal L et al. Genome-wide association study of 21,000 cases and 95,000 controls identified new risk loci for atopic dermatitis. *Nat Genet* 2015.
- Chang ALS et al. Assessment of the genetic bases of rosacea by genome-wide association study. *J Invest Dermal* 2015.
- Nakashima M et al. *Nat Genet* 2010 (GWAS keloid; four susceptibility loci).
- Sun Q et al. Genome-wide association study of hidradenitis suppurative. 2023 (SOX9, KLF5).
- Ferwerda B et al. Human dectin-1 deficiency and mucocutaneous fungi infections. *N Engl J Med* 2009.
- Jin Y et al. Genome-wide association studies of autoimmune vitiligo identify 23 new risk loci. *Nat Genet* 2016.
- Privé F et al. *Am J Hum Genet* 2022 (PGS001871); Shen M et al. *J Am Acad Dermatol* 2022 (PGS002293); Al-Janabi A et al. *J Invest Dermatol* 2023 (PGS003486); Roberts GHL et al. *Am J Hum Genet* 2019 (PGS000738).
- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

Generated in 02-10-2026 (local time) by the scripts of this report.